

Exploratory Data Analysis in Power BI

Exploratory Data Analysis

What is EDA?

Exploratory Data Analysis (EDA) is an approach to **analysing and visualising data to summarise** the main characteristics, often with the help of statistical graphics and other data visualisation methods.

The **primary goal of EDA** is to **uncover patterns, relationships, and trends** in the data, as well as to identify any **outliers** or **anomalies**.

It is a **crucial step in data analysis and science processes** and is typically performed before formal statistical modelling.

Why is EDA important?

Exploratory Data Analysis (EDA) helps us to **understand our data** without making any assumptions.

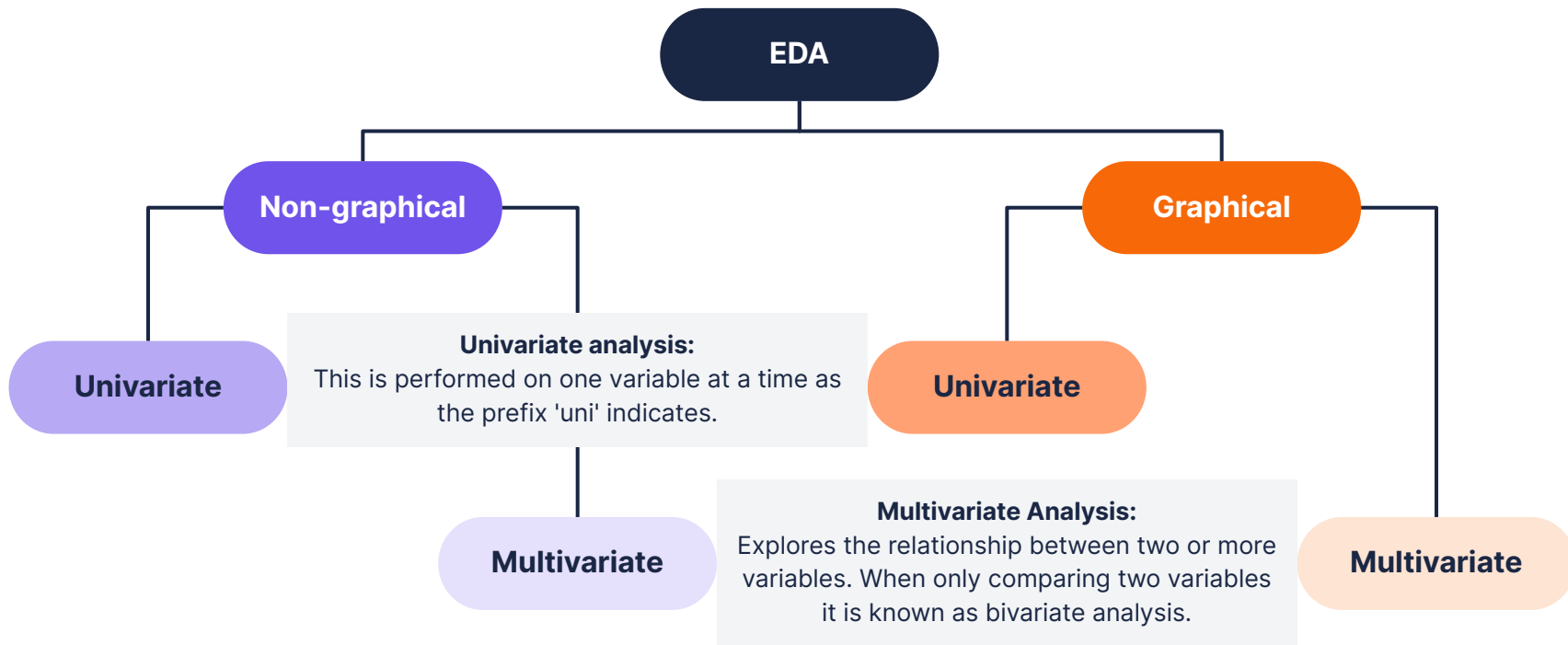


Before we can gather insights or build a model, we first need to understand our data. EDA provides **context** and **guidance** on the course of action to take and will **assist in interpreting** our results correctly.

Without doing EDA we will not understand our data or insights fully.

The different types of EDA

EDA is generally classified in two ways: **non-graphical** or **graphical**, and **univariate** or **multivariate** analysis.

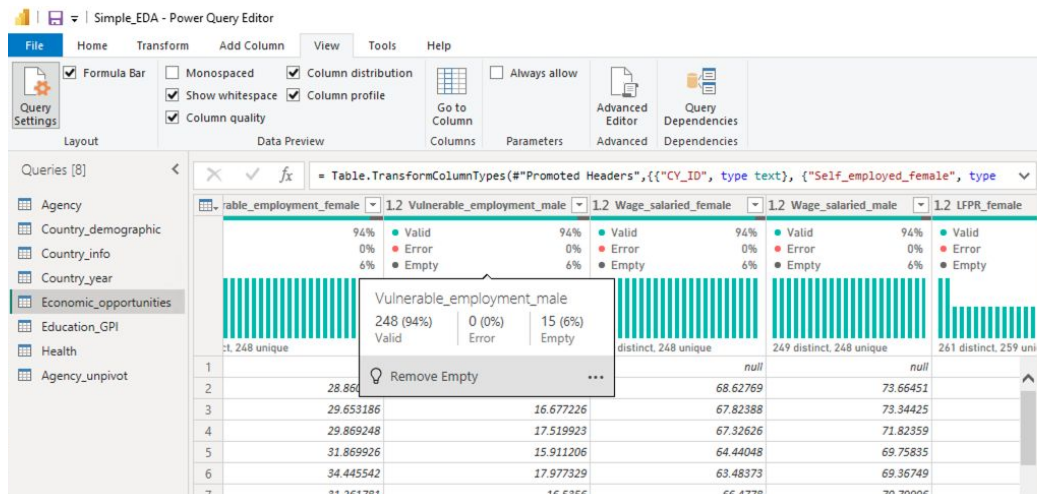


Initial EDA

Before we jump into graphical and non-graphical univariate and multivariate analyses, an initial EDA phase would include considering the **data we have**, their **types** and **units**, and the **number of missing values** we are dealing with.

Both **graphical and non-graphical approaches** would include this step. We need a very **basic understanding of the data** we have before considering slightly more advanced techniques.

In **Power BI**, we can use **implicit or explicit measures** or the **Power Query Editor** to analyse erroneous and missing values, as well as the distribution.



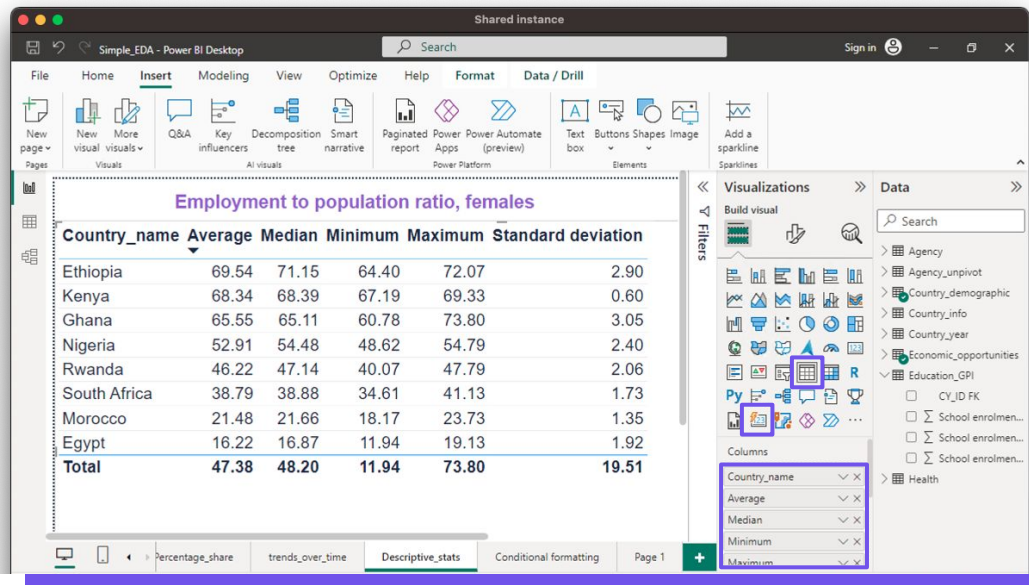
Initial EDA in Power BI using the **Power Query Editor**.

Non-graphical univariate analysis

Non-graphical univariate analysis provides a **quantitative summary** of the **distribution** and **characteristics** of individual variables.

These statistics are useful for gaining insights into the **central tendency**, **variability**, and **shape** of the data distribution. While graphical methods can provide a visual representation, descriptive statistics offer **numerical values that can be easily compared and communicated**.

In **Power BI**, we can use **implicit measures** to create a quantitative summary and a **table** or **card** visualisation to communicate the results.



Non-graphical multivariate analysis

Non-graphical multivariate analysis involves **examining relationships between two or more variables** without relying on visualisations. **Correlation analysis is a key technique** used in this context to understand the strength and direction of relationships between variables.

Correlation analysis is a powerful tool for **assessing relationships** between variables, but it's important to note that correlation does not capture all types of relationships (e.g. non-linear relationships). Other multivariate techniques, such as **regression analysis**, **factor analysis**, and **covariance analysis**, may be employed in non-graphical multivariate analysis depending on the specific research or analytical objectives.

In **Power BI**:

- **Correlation analysis:** We can use DAX (Data Analysis Expressions) to calculate correlation coefficients.
- **Regression and factor analysis:** Power BI doesn't natively have these but we can use R or Python scripts to perform more advanced statistical analyses.
- **Covariance analysis:** We can use DAX (Data Analysis Expressions) to calculate covariance and covariance matrices.

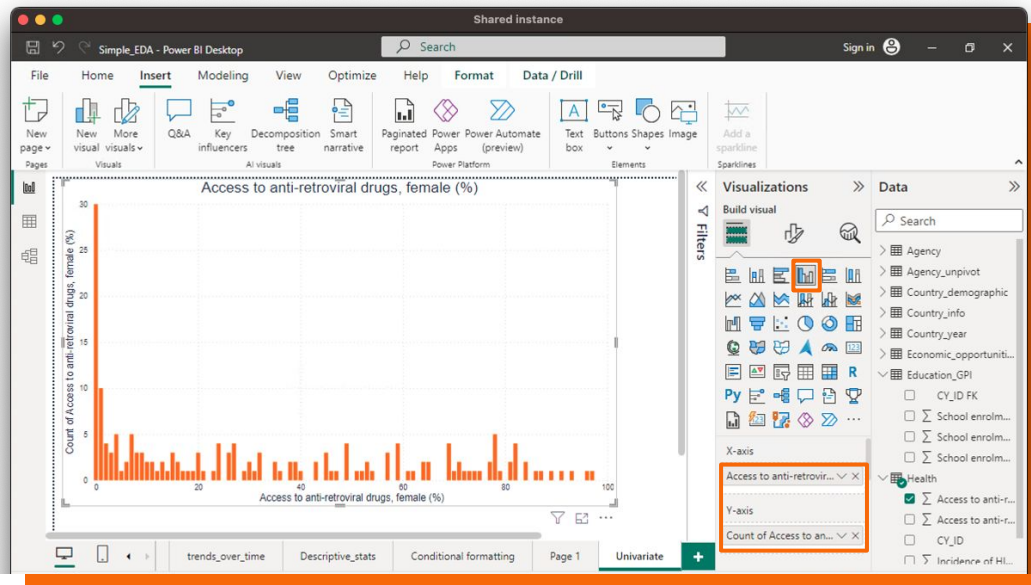
Graphical univariate analysis

Graphical univariate analysis involves using **visualisations** to **explore** and **understand** the **distribution** and **characteristics** of **individual variables**.

We need to **choose visualisations that are appropriate** for the type of variable we are analysing (numerical or categorical) and the specific insights we are seeking.

Experimenting with different chart types can help us **uncover patterns and outliers in the data**.

In **Power BI**, we can create a histogram by using the **same column in both the x and y-axis wells**, and setting the y-axis to the **count** of that column.

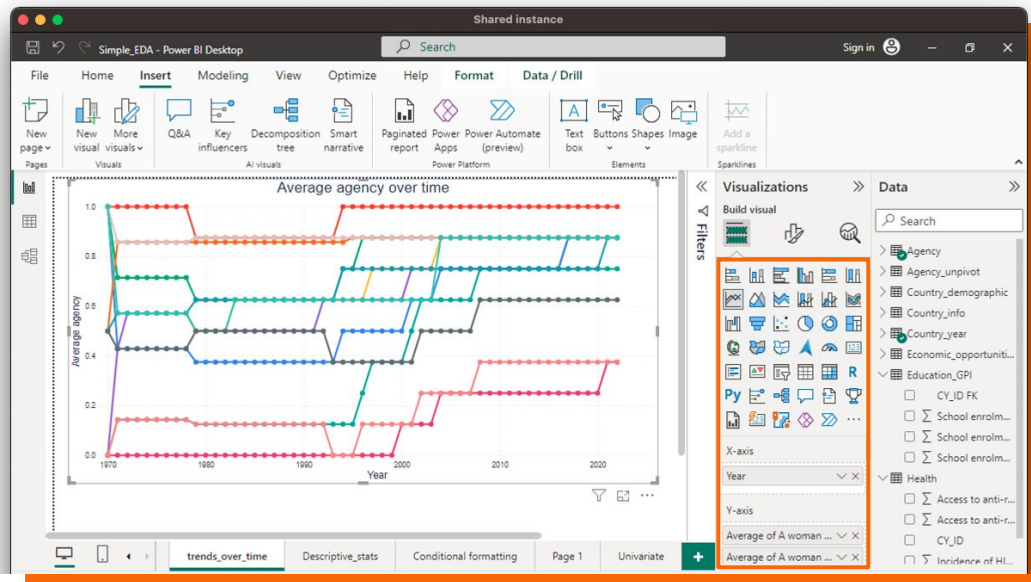


Graphical multivariate analysis

Graphical multivariate analysis involves using **visualisations to explore relationships** between two or more variables. We can use visualisations such as **heatmaps**, **scatter plots**, and **pair plots** to investigate the relationship.

These visualisations help us **explore patterns, correlations, and trends** in the data across multiple dimensions. Selecting the most appropriate visualisation **depends on the nature of the variables** and the **specific questions** we aim to answer.

In **Power BI**, we can use many different visualisations, from **line charts to bubble charts and treemaps**.



EDA in Power BI

While the specific techniques and visualisations we use will **depend on the nature of the data** and the **questions we are trying to answer**, Power BI also offers a **wide range of tools and features** to support a **comprehensive EDA process**.

Microsoft is also **constantly adding new tools and features to Power BI**, driven by the advancements in AI, that improves our ability to **automate** some of the EDA process.

However, it's important to note that the true value of EDA is **gaining deep insight and understanding of data**, which very often will require our intervention.

Additional **Power BI Desktop tools and features** include:

- **Key influencers** visualisation
- **Analyze** feature
- **Q&A** visualisation
- **Conditional formatting** in a *table* visualisation

Power BI service also includes several additional EDA tools such as detecting data anomalies, trends, etc. through the *Quick insights* feature.

